dbt Snowflake Project to Master dbt Fundamentals in Snowflake

Business Overview:
DBT (Data Build Tool) is a popular open-source command-line tool used in the data transformation and modeling process. It is widely used in data warehousing and business intelligence projects for building data pipelines, transforming data, and building data models.

DBT is designed to help data analysts and engineers create, maintain, and test modular SQL scripts that can be run sequentially to extract data from various sources, transform it into a structured format, and load it into a target system. It uses the concept of "models" to organize and define SQL queries that transform data from source to target. You can define dependencies between models and run tests to ensure the accuracy of the transformed data.

There are several reasons **why the Data Build Tool (DBT) has become popular and in demand** in the data engineering and analytics market:

- **Open-source:** DBT is an open-source tool, which means it is free to use, and its source code is available for anyone to modify and improve upon. This has made it accessible to a wide range of users, from small startups to large enterprises.
- **Data transformation and modeling:** DBT provides a powerful and easy-to-use framework for data transformation and modeling. It allows users to define data models using SQL, which can be used to transform and manipulate data from various sources into a structured format.
- **Scalability:** DBT is designed to be highly scalable, making it ideal for data warehousing and business intelligence projects of all sizes. It can be used to build and maintain large data pipelines with ease.
- **Integration with data warehouses and BI tools:** DBT can be integrated with popular data warehouses such as Snowflake, Redshift, and BigQuery, and can be used with various data visualization tools such as Tableau and Looker. This makes it easier for users to integrate it into their existing data infrastructure.
- **Testing and documentation:** DBT provides built-in testing and documentation features, making it easier for users to ensure the accuracy of their data pipelines and maintain documentation for future reference.

DBT (Data Build Tool) and Snowflake are two powerful tools that can be used together for data transformation and modeling. Snowflake is a cloud-based data warehousing platform that allows users to store, manage, and analyze large amounts of data. DBT, on the other hand, is an open-source tool that helps users transform and model data using SQL. DBT and Snowflake can be integrated seamlessly, with DBT providing the necessary data transformation and modeling capabilities to enhance Snowflake's functionality. DBT is specifically designed to work with data warehouses like Snowflake, making it easy to create data pipelines and perform transformations on Snowflake data. Additionally, DBT's built-in testing and documentation features can help ensure the accuracy of Snowflake data pipelines and maintain documentation for future reference.

Aim:

This project series aims to provide a comprehensive understanding of the DBT (Data Build Tool). The first project in the series will cover the fundamental aspects of DBT, including the need for the tool and where it fits in the ETL process. We will explore data modeling and logging, which are essential aspects of DBT. To demonstrate how DBT can be used in real-life scenarios, we will be using Netflix movies and series data stored in Snowflake. By using DBT with Snowflake, we will show how businesses can leverage these tools to solve complex data problems. Through this project, we will gain insight into the various features of DBT and how they can be used to build robust data pipelines. With this knowledge, users can learn to optimize their data transformation and modeling workflows, making them more efficient and effective for their organization's needs.

Tech Stack

➔
Language: Python, SQL

➔
Tool: DBT

➔
Database: Snowflake

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Services: AWS EC2

Key Takeaways

- Understanding the DBT in detail
- Understanding the need for DBT
- Understanding the fundamentals of DBT
- Installing DBT on the Windows system
- Installing DBT on the EC2 instance

- Creating DBT project
- Understanding the directory structure of the DBT project
- Uploading raw data to Snowflake using Python script
- Understanding the Data Modelling in DBT
- Importance of profiles file in the DBT project
- Understanding the importance of logging in DBT
- Use of Materializations in DBT
- Creating Variables for the DBT project
- Sources, Prehook, and Posthook in the DBT project
- Creating different models to answer business questions
- Executing data models in DBT
- Different types of testing in DBT
- Difference between DBT Cloud VS DBT Core

Architecture Diagram: